Lorenzo Minervino

Software Engineer — Backend & Distributed Systems — Python, Java, TypeScript

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Professional Summary

Junior Software Engineer with a backend focus, combining academic and personal project experience in Python, Java and TypeScript with a strong foundation in distributed systems. Built REST APIs, relational data models, asynchronous data pipelines and fault-tolerant distributed services using Spring Boot, Node.js, PostgreSQL, MySQL, RabbitMQ and Docker. Also applies Rust for systems-level distributed programming and has hands-on applied-AI experience, including neural network training and Python/Rust integration via PyO3. Computer Engineering candidate at the University of Buenos Aires — all coursework completed, thesis defense pending, expected 2026. Currently working as an AI Trainer, evaluating technical programming and mathematics responses.

Technical Skills

Programming Languages: Python, Java, TypeScript/JavaScript, Rust, C
Backend: REST APIs, Spring Boot, Node.js (NestJS), JWT Authentication, Microservices
Databases: PostgreSQL, MySQL, MongoDB
Distributed Systems: RabbitMQ, Replication, Fault Tolerance, Leader Election, Consistency Models, Replicated Logs
AI & Data: Neural Network Training, Data Processing Pipelines, PyO3 (Rust/Python Integration), AI-assisted Development
Systems & DevOps: Linux, Docker, Docker Compose, Git/GitHub, GitHub Actions, CI/CD Fundamentals, TCP Networking, Concurrency

Selected Projects

Oxidized Neural Orchestra — Final Degree Project · Rust, Python (PyO3), Docker · team of 3
github.com/lminervino18/oxidized-neural-orchestra

- Distributed system for training neural networks across a cluster of nodes, implementing parameter server, ring all-reduce, and a hybrid strategy that promotes workers to parameter servers once gradients converge.
- Designed as a Cargo workspace with a custom TCP communication layer, a runtime-configurable node binary, Python bindings via PyO3, a live-monitoring terminal UI, and Docker Compose for multi-node cluster simulation.

Distributed Transaction Analysis Pipeline — Python, RabbitMQ, Docker · team of 3
github.com/lminervino18/distribuidos-tp-money-laundering-analysis

- Multi-stage data pipeline (filter, join, group-by, aggregate) with asynchronous processing distributed across RabbitMQ queues, coordinated via a ring protocol for end-of-stream detection.
- Fault tolerance built on leader/replica supervision, heartbeats, checkpointing and automatic node recovery, validated through chaos testing that kills and restores random nodes.

CONICET Researchers Platform — Java, Spring Boot, Spring Data JPA, MySQL, React/TypeScript · team of 2
github.com/lminervino18/researchers-page-conicet

- REST API built with Spring Boot and Spring Data JPA for publishing scientific work, team information and outreach content, backed by relational data modeling in MySQL.
- Implemented JWT-based authentication and integrated a React/TypeScript frontend consuming the API endpoints.

Rustic Airlines — Rust, Docker · team of 4
github.com/lminervino18/rustic-airlines

- Distributed flight management system on a Cassandra-inspired database engine, supporting multi-node replication, configurable consistency levels, fault tolerance and cluster growth via seed-node discovery.

Professional Experience

AI Trainer — Outlier — Remote · Jan 2025 – Present

- Evaluated AI-generated responses to programming and mathematics problems for technical correctness, reasoning and clarity.
- Delivered structured feedback documenting each error and the reasoning behind the correction.

Education

Computer Engineering — University of Buenos Aires (UBA)
All coursework completed. Thesis defense pending — Expected 2026.

Languages

Spanish: Native · English: Upper-Intermediate (B2), Cambridge First Certificate (FCE)